

Validity Analysis: DRBENCHMARK

Target context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Overall risk: HIGH

Deployment Context

Use case: In a hospital setting, a clinician uses software to manage unstructured French clinical cases by applying a model evaluated on part-of-speech tagging, named-entity recognition, and multi-label classification. The system identifies specific clinical and temporal entities while mapping pathologies to standardized medical classification chapters and axes to automatically generate a patient’s clinical profile. These outputs directly help the clinician summarize complex medical histories and prioritize care based on the severity of identified symptoms, ensuring that life-critical decisions are supported by accurately synthesized data.

Target population: French-speaking medical professionals, including clinicians, hospital physicians, and nursing staff in French-speaking health systems.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology 	1	Serious concern	high
Output Content 	2	Significant gaps	high
Output Form 	3	Moderate gaps	medium
Average	2.3		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

DrBenchmark is a well-engineered, broadly scoped French biomedical NLU benchmark that provides reproducible evaluation infrastructure, formal-French biomedical text consistency, and partial coverage of pharmaceutical vocabulary. For the specific deployment of French Pharmaceutical Regulatory Affairs document compliance NLP, however, the benchmark has

fundamental validity gaps on every HIGH-priority dimension. Output Ontology (OO) is the most critical: the STS scoring rubric (0–5 general semantic proximity) empirically fails to distinguish legally consequential small-magnitude differences (omission of safety qualifiers, dose-threshold variations) that trigger formal Type IA/IB/II variation submissions under EU/ANSM frameworks, and NER label spaces (UMLS Semantic Groups, ICD-10 chapters, MeSH axes) do not encode regulatory ontology categories (INN, ATC, excipient, contraindication qualifier, MedDRA PT). Input Ontology (IO) and Input Content (IC) reflect this same gap structurally — no task targets SmPC/PIL/CTD compliance checking; QUAERO EMEA is the only regulatory-genre source and is used only for general NER. Output Content (OC) lacks regulatory affairs annotation authority across all datasets, with concrete annotation-judgment artifacts visible (e.g., DEFT2020 STS scores treating safety-qualifier omissions as highly similar). Input Form (IF) is well-matched (text-only French, Latin alphabet, HuggingFace-compatible). Output Form (OF) is structurally compatible but lacks confidence calibration and human-review threshold benchmarking. Overseas territory tropical pathology vocabulary is absent across all datasets despite being identified as a secondary deployment priority.

Practical Guidance

What This Benchmark Measures

DrBenchmark measures general French biomedical NLU competence across NER, POS, classification, MCQA, and STS in clinical and research genres. For the regulatory deployment, it most strongly probes (a) Input Form fitness — whether models handle formal written French biomedical text (well-covered, Score 4); (b) baseline pharmaceutical vocabulary recognition through QUAERO EMEA NER, PxCorpus 38-class posology NER, and DEFT-2021 SUBSTANCE/DOSAGE NER; and (c) baseline French biomedical STS scoring via CLISTER and DEFT2020. It provides useful screening signal for whether a candidate model can process French pharmaceutical text at all.

Construct Depth

Construct depth is shallow relative to the regulatory compliance construct. The benchmark probes general semantic proximity but not regulatory-legal equivalence (Output Ontology Score 1). It tests entity recognition over clinical/research ontologies but not regulatory entity ontologies (Output Ontology, Input Content). Annotation authority is clinical/research, not regulatory (Output Content Score 2). The benchmark therefore provides necessary-but-far-from-sufficient evidence: a model that fails on QUAERO EMEA or DEFT2021 NER is likely unfit for regulatory deployment, but strong benchmark performance does NOT establish fitness for compliance checking, equivalence judgment, or version-comparison tasks.

What Else You Need

Substantial supplementation is required: (1) a regulatory-calibrated STS evaluation set with sentence pairs labeled by regulatory affairs specialists for legal-equivalence judgments aligned to variation type IA/IB/II thresholds [WEB-6] — addresses the OO and OC gaps; (2) a regulatory NER evaluation with INN, ATC, excipient, contraindication qualifier, and MedDRA PT label categories — addresses OO and IC gaps and would benefit from a French-language analogue to ADE Eval [WEB-12]; (3) SmPC/PIL/CTD genre-specific test sets for both NER and STS — addresses IO and IC; (4) confidence calibration evaluation with documented score thresholds for human-review escalation — addresses OF; (5) overseas territory tropical pathology vocabulary supplementation if DOM scope is operationalized; (6) regulatory-affairs-expert re-annotation of a representative DrBenchmark sample to quantify clinical-vs-regulatory annotator divergence.